Lecture Scribe Conversion

L4_A.pdf
PDF

You are a course expert in computation probability and act as a scholar.
I am providing you lecture PowerPoint presentation. All you have to do is to convert it into a proper and a detailed version of it in lecture scribe format. The lecture scribe should contain:
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Include derivation steps
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Include all the applications given in the ppt
Split down the complex concepts into step by step and simpler explanations
Do explain using conceptual approach and mathematic approach
If you find any inconsistency, solve it and generate the most accurate explanation
Give summary and make sure to include important topics
Give the formulas sheet
Do not assume prior knowledge
Do not skip any slide and topic

Great—I've carefully gone through your uploaded lecture slides **L4_A.pdf** (20 pages) and converted them into a **clean, detailed, student-friendly lecture scribe**. I'm following your instructions strictly: no assumed prior knowledge, every slide/topic covered, concepts explained in simple words, logic + math both, examples included, and a summary + formula sheet at the end. The content below is based directly on the slides you shared .

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
(From slides dated Jan 15, 2026)

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1. Motivation: Why Do We Study Probability in Computing?

Engineering Applications (from slides 2–6)

1. Speech Recognition System

In speech recognition, a computer listens to a sound signal and tries to **match it with stored templates** like “Hello”, “Yes”, “No”, “Bye”.

- The spoken signal is a **time signal**: x(t)
- The system compares it with stored **templates** for different words and speakers.
- Because of:
  - Noise
  - Different speakers (male, female, child)
  - Distortions
    the same word never looks exactly the same.

So the system does not decide with certainty. Instead, it asks:
Which word has **higher probability** given this observed signal?

This is a **probability decision problem**.

---

2. Radar System

Radar sends a signal and receives a reflected signal.

Two hypotheses:

- H0: Only noise is received (no target)
- H1: Signal + noise is received (target present)

Let:

- Y = received signal

- W = noise
- S = actual signal

Then:

- Under H0: Y = W
- Under H1: Y = S + W

Decisions can be:

- Correct detection
- False alarm
- Miss detection

Probabilities like:

- P(False Alarm)
- P(Miss Detection)

must be **computed and optimized** → again, probability is essential.

---

3. Communication Networks

In networks:

- Packets travel from source to destination
- There is delay, loss, congestion
- We care about:
  - Probability that a packet arrives
  - Probability of delay exceeding some limit (QoS: Quality of Service)

So again, **uncertainty → probability modeling**.

---

2. Outline of the Theory (from slides 7–8)

We study:

1. Experiments, Sample Space, Events
2. Axioms of Probability
3. Corollaries and Propositions
4. How to assign probability
5. Joint Probability (later)
6. Conditional Probability (later)

This lecture starts with the **foundations**.

---

3. Basic Definitions: Experiments, Outcomes, Events, Sample Space

3.1 Experiment (E) (slide 9)

Definition:
An experiment is a **procedure** that produces some result.

Example:
Tossing a coin 5 times. Call this experiment E_5.

This does not mean one outcome; it means the **process** of doing the tosses.

3.2 Outcome (ξ) (slide 10)

Definition:
An outcome is a **possible result** of an experiment.

Example:
For 5 coin tosses, one possible outcome is:
ξ_1 = H T H T H

This is **one specific sequence**.

3.3 Event (slide 11)

Definition:
An event is a **set of outcomes** of an experiment.

Example:
Let C = event that “number of heads is even” in 5 tosses.

Then C includes many outcomes, such as:

- H H T T T (2 heads)
- H H H H T (4 heads)

- T T T T T (0 heads)

  etc.

So:
Event = collection of outcomes satisfying some condition.

3.4 Sample Space (S) (slide 12)

Definition:
The sample space S is the **set of all possible distinct outcomes** of an experiment.

Important properties:

- Mutually Exclusive: You can get **only one** outcome at a time.
- Collectively Exhaustive: **All** possible outcomes are included.

Example:
Flip one coin:
S = {H, T}

You cannot get both at once, and you cannot get anything else.

3.5 Types of Sample Spaces (slides 13–14)

The sample space S can be:

1. Discrete
   - Finite or countably infinite

     Examples:
   - One coin toss: {H, T}
   - One die roll: {1,2,3,4,5,6}
   - Flip coin until first tail: {T, HT, HHT, HHHT, ...} (countably infinite)
2. Continuous
   - Values come from a continuous interval

     Example:
   - Random number in [0, 1]
   - Time taken by a process (any real value in an interval)

3.6 Examples from the Slide (slide 14)

- Flipping a fair coin once → Discrete, finite
- Rolling a die → Discrete, finite
- Rolling two dice → Discrete, finite (36 outcomes)
- Flip coin until tails occurs → Discrete, countably infinite
- Random number in [0,1] → Continuous

---

4. Probability and Axioms (slide 15)

4.1 What is Probability?

Definition:
Probability is a **number** that measures how likely an event is.

Formally:
It is a function Pr(A) that assigns a number to each event A.

4.2 Axioms of Probability

These are **basic rules** that probability must follow.

Axiom 1:
For any event A:
$0 \leqslant Pr(A) \leqslant 1$

Meaning:
Probability is never negative and never more than 1.

Axiom 2:
If S is the sample space:
$Pr(S) = 1$

Meaning:
Something in the sample space **must happen**.

Axiom 3 (Additivity):

(a) If A and B are disjoint ($A \cap B = \varnothing$), then:
$Pr(A \cup B) = Pr(A) + Pr(B)$

(b) For infinitely many mutually exclusive events $A_1, A_2, A_3, \ldots$:
$Pr(\cup_i A_i) = \Sigma_i Pr(A_i)$

Meaning:
If events cannot happen together, the probability of "one of them happening" is the **sum** of their probabilities.

---

5. Corollary from Axioms (slides 16–17)

For a **finite** number M of mutually exclusive events $A_1, A_2, \ldots, A_M$:

$Pr(\cup_{i=1}^{M} A_i) = \Sigma_{i=1}^{M} Pr(A_i)$

This is just the **finite version** of Axiom 3.

Important note from slide:

- Axiom 3 is needed especially when the sample space is **infinite**.

---

6. Propositions from Axioms (slides 18–20)

Proposition 2.1

$Pr(A^c) = 1 - Pr(A)$

Explanation:

- A^c means "A does not happen"
- Either A happens or A does not happen
- Total probability = 1
  So:
  $Pr(A) + Pr(A^c) = 1$
  $\Rightarrow Pr(A^c) = 1 - Pr(A)$

Proposition 2.2

If $A \subset B$, then:

$\Pr(A) \leqslant \Pr(B)$

Explanation:

- If every outcome in A is also in B, then B is a **bigger event**
- Bigger event cannot have smaller probability than a smaller part of it

So probability respects **set inclusion**.

---

7. Conceptual + Mathematical View (Very Important)

Conceptually:

- Experiment = process
- Outcome = one result
- Event = set of results
- Sample space = all possible results
- Probability = how likely an event is

Mathematically:

- Events are sets
- Sample space S is the universal set
- Probability $\Pr(\cdot)$ is a function:
  $\Pr: \text{Events} \to [0,1]$
- Axioms define how this function must behave

---

8. Summary of This Part of the Lecture

You learned:

- What is an experiment, outcome, event, sample space
- Types of sample spaces: discrete, countable, continuous
- What probability means
- Three axioms of probability
- Corollary for finite mutually exclusive events
- Two important propositions:
  - Complement rule: $\Pr(A^c) = 1 - \Pr(A)$
  - Monotonicity: If $A \subset B$, then $\Pr(A) \leqslant \Pr(B)$

These are the **foundations** needed before studying:

- Joint Probability
- Conditional Probability

---

9. Formula Sheet (From This Lecture Part)

- $0 \leqslant \Pr(A) \leqslant 1$
- $\Pr(S) = 1$
- If $A \cap B = \varnothing$:
  $\Pr(A \cup B) = \Pr(A) + \Pr(B)$
- For mutually exclusive $A_1, A_2, \ldots$:
  $\Pr(\bigcup_i A_i) = \sum_i \Pr(A_i)$
- Complement:
  $\Pr(A^c) = 1 - \Pr(A)$
- If $A \subset B$:
  $\Pr(A) \leqslant \Pr(B)$

If you want, next we can continue **the same way** for:

- Joint Probability (with full derivations and examples)
- Conditional Probability (cards, poker, missing key, etc.)
  just upload the next part or say “continue from joint probability”.

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Perfect—I've carefully studied your uploaded slides **L4_A.pdf** (20 pages) and converted **every slide and topic** into a clean, structured, **lecture-scribe style** explanation, exactly following your instructions: simple language, step-by-step logic, conceptual + mathematical views, derivations where applicable, examples, applications, summary, and a formula sheet. The content below is derived directly from your slides .

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
(Date: January 15, 2026)

Slide 1: Lecture Introduction

Topic: Joint Probability and Conditional Probability

This lecture is part of CSE400 (Fundamentals of Probability in Computing).
Before studying joint and conditional probability, the lecture first builds **foundations**:

- What is probability?
- What are experiments, outcomes, events, and sample spaces?
- What rules (axioms) does probability follow?

These basics are necessary because **joint** and **conditional** probability are built on them.

Slide 2: Why Should We Learn CSE400?

Main idea: Probability is used everywhere in real systems.

Examples mentioned:

- Daily life reasoning (we constantly judge chances)
- Engineering applications:
  1. Speech Recognition
  2. Radar Systems
  3. Communication Networks

Meaning:
In real systems, we never have perfect certainty. We always deal with **uncertainty**, so we need **probability** to model, measure, and make decisions.

Slides 3–4: Engineering Application 1 – Speech Recognition System

Conceptual Explanation

A speech recognition system:

- Has a **vocabulary** (Hello, Yes, No, Bye, etc.)
- For each word, it stores **templates** (example sound patterns)
- When a person speaks, the system records a signal x(t)
- This signal is compared with stored templates

Problem:

- Different speakers (male, female, child)

- Noise and disturbances
- Same word never looks exactly the same

So the system asks:
Which word is **most probable** given the observed signal?

In short:
We are not doing exact matching → we are doing **probabilistic decision making**.

This is why probability is essential in speech recognition (and NLP in general).

Slide 5: Engineering Application 2 – Radar System

Physical Model

Radar sends a signal and receives something back.

Two hypotheses:

- H_0: No target → received signal Y = W (only noise)
- H_1: Target present → received signal Y = S + W (signal + noise)

Where:

- S = true signal
- W = noise
- Y = received signal

Decision Outcomes

Possible cases:

- Correct detection
- False alarm (detect target when there is none)
- Miss detection (fail to detect target)

We define probabilities like:

- P(False Alarm)
- P(Miss Detection)

These probabilities guide **system design and performance**.

Again: decisions are based on **probability**, not certainty.

Slide 6: Engineering Application 3 – Communication Networks

In networks:

- Data packets travel from source to destination
- There is delay, loss, congestion
- Quality of Service (QoS) depends on:
  - Probability of packet arrival
  - Probability of delay being too large
  - Probability of failure

So we model:

- Arrival of packets
- Delay (latency)
- Throughput

All of these are **random** → need probability theory.

Slides 7–8: Lecture Outline

Three main parts:

1. Introduction to Probability Theory
   - Experiments, Sample Space, Events
   - Axioms of Probability
   - Corollaries and Propositions
   - How to assign probability
2. Joint Probability
   - Motivation, notation, concepts
   - Examples: Card deck, Costume party
3. Conditional Probability
   - Motivation, notation, concepts
   - Examples: Cards without replacement, Poker, Missing key

In this file, the slides mainly cover **Part 1: Foundations**.

Part 1: Introduction to Probability Theory

Slide 9: Experiment

Definition:
An experiment (E) is a **procedure** we perform that produces some result.

Example:
Tossing a coin 5 times → call this experiment E_5.

Important:

- Experiment = process
- Not a single result, but the **action** that generates results

---

Slide 10: Outcome

Definition:
An outcome (ξ) is a **possible result** of an experiment.

Example:
For 5 coin tosses, one outcome is:
ξ_1 = H T H T H

So:

- Experiment: Toss coin 5 times
- Outcome: One specific sequence like HHTHT, TTTTT, etc.

Slide 11: Event

Definition:
An event is a **set of outcomes** of an experiment.

Example:
Let C be the event:
"Number of heads is even" in 5 tosses.

Then C contains many outcomes:

- 0 heads
- 2 heads
- 4 heads

So:

Event = collection of outcomes that satisfy some condition.

Slide 12: Sample Space

Definition:

The sample space (S) is the **set of all possible outcomes** of an experiment.

Properties:

- Mutually exclusive: You can get only one outcome at a time
- Collectively exhaustive: All possible outcomes are included

Example:

Flip one coin:

S = {H, T}

You cannot get anything else, and you cannot get both at once.

Slides 13–14: Types of Sample Spaces

The sample space S can be:

1. Discrete
   - Finite or countably infinite

     Examples:
   - One coin toss: {H, T}
   - One die roll: {1,2,3,4,5,6}
   - Flip coin until first tail: {T, HT, HHT, HHHT, ...} (countably infinite)
2. Continuous
   - Values come from a continuous interval

     Examples:
   - Random number in [0, 1]
   - Time taken by a process

Examples listed in slides:

- Flipping a fair coin once → Discrete
- Rolling a die → Discrete
- Rolling two dice → Discrete

- Flip coin until tails occurs → Discrete, countably infinite
- Random number in [0,1] → Continuous

Slide 15: Probability and Axioms

What is Probability?

Definition:
Probability is a **number** that measures how likely an event is.

Formally:
It is a function Pr(A) that assigns a number to each event A.

Axioms of Probability

These are **basic rules** that do not need proof.

Axiom 1:
For any event A:
$0 \leqslant Pr(A) \leqslant 1$

Meaning:
Probability is always between 0 and 1.

Axiom 2:
If S is the sample space:
$Pr(S) = 1$

Meaning:
Something in the sample space **must happen**.

Axiom 3 (Additivity):

(a) If $A \cap B = \varnothing$ (disjoint events):
$Pr(A \cup B) = Pr(A) + Pr(B)$

(b) For infinitely many mutually exclusive events $A_1, A_2, A_3, \ldots$:
$Pr(\cup_i A_i) = \Sigma_i Pr(A_i)$

Meaning:
If events cannot occur together, the probability of “any one of them happening” is the **sum** of their probabilities.

Slides 16–17: Corollary from Axioms

For a **finite** number M of mutually exclusive events $A_1, A_2, \ldots, A_M$:

$Pr(\cup_{i=1}^{M} A_i) = \Sigma_{i=1}^{M} Pr(A_i)$

Explanation:
This is just the **finite version** of Axiom 3.

Note from slide:

- Axiom 3 is especially important when the sample space is **infinite**.

Slides 18–20: Propositions from Probability Axioms

Proposition 2.1: Complement Rule

$Pr(A^c) = 1 - Pr(A)$

Derivation (logic):

- Either A happens or A does not happen
- These two cover the entire sample space
  So:
  $Pr(A) + Pr(A^c) = Pr(S) = 1$
  $\Rightarrow Pr(A^c) = 1 - Pr(A)$

Proposition 2.2: Monotonicity

If $A \subset B$, then:
$Pr(A) \leqslant Pr(B)$

Logic:

- A is a part of B
- B includes all outcomes of A and possibly more
- So B cannot be less likely than A

Conceptual + Mathematical View (Important)

Conceptually:

- Experiment = process
- Outcome = one result
- Event = set of results
- Sample space = all results
- Probability = how likely an event is

Mathematically:

- Events are sets
- S is the universal set
- $Pr(\cdot)$ is a function: Events $\rightarrow$ [0, 1]
- Axioms define how this function behaves

Summary of This Part of the Lecture

You learned:

- Experiment, Outcome, Event, Sample Space
- Types of sample spaces: Discrete and Continuous
- What probability means
- Three axioms of probability
- Corollary for finite mutually exclusive events
- Two key propositions:
  - $Pr(A^c) = 1 - Pr(A)$
  - If $A \subset B$, then $Pr(A) \leqslant Pr(B)$

These are the **foundation** for:

- Joint Probability
- Conditional Probability (which come next in the lecture)

Formula Sheet (From These Slides)

- $0 \leqslant \Pr(A) \leqslant 1$
- $\Pr(S) = 1$
- If $A \cap B = \varnothing$:

  $\Pr(A \cup B) = \Pr(A) + \Pr(B)$
- For mutually exclusive A_i:

  $\Pr(\bigcup_i A_i) = \sum_i \Pr(A_i)$
- Complement:

  $\Pr(A^c) = 1 - \Pr(A)$
- If $A \subset B$:

  $\Pr(A) \leqslant \Pr(B)$

If you want, I can now continue in **the same detailed lecture-scribe style** for:

- Joint Probability (with card deck & costume party examples), and
- Conditional Probability (cards, poker, missing key).

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Lecture 4: Joint Probability and Conditional Probability
Instructor: Dhaval Patel, PhD
(Date: January 15, 2026)

Slide 1: Lecture Introduction

Topic: Joint Probability and Conditional Probability

This lecture belongs to the course **CSE400: Fundamentals of Probability in Computing**.
Before learning **joint** and **conditional** probability, the lecture first builds the **basic foundation**:

- What probability is
- What experiments, outcomes, events, and sample spaces are
- What rules (axioms) probability must follow

These basics are necessary because all advanced probability topics are built on them.

Slide 2: Why Should We Learn CSE400?

Main idea: **Real systems are uncertain**, so we use probability.

Examples mentioned:

- Daily life reasoning (we often say “probably”, “maybe”, “chance is high”)
- Engineering applications:
  1. Speech Recognition
  2. Radar Systems
  3. Communication Networks

Meaning:
In real engineering systems, decisions are not perfect or exact. We **model uncertainty using probability**.

Slides 3–4: Engineering Application 1 – Speech Recognition System

Conceptual View

A speech recognition system:

- Has a **vocabulary** (Hello, Yes, No, Bye, etc.)
- For each word, it stores **templates** (example sound waveforms)
- When a person speaks, the system records a signal $x(t)$
- This signal is compared with stored templates

Problems:

- Different speakers (male, female, child)
- Noise and interference
- Same word never looks exactly the same

So the system does **not** ask:
“Is this exactly equal to template A?”

Instead, it asks:
“Which word is **most probable** given this signal?”

This is a **probabilistic decision problem** (also related to NLP and signal processing).

Slide 5: Engineering Application 2 – Radar System

Physical Model

Radar sends a signal and receives something back.

Two hypotheses:

- H_0: No target → received signal $Y = W$ (only noise)
- H_1: Target present → received signal $Y = S + W$ (signal + noise)

Where:

- S = true signal
- W = noise
- Y = received signal

Decision Outcomes

Possible decisions:

- Correct detection
- False alarm (detect target when there is none)
- Miss detection (fail to detect target)

We define probabilities like:

- $P(\text{False Alarm})$
- $P(\text{Miss Detection})$

System performance is judged using these probabilities → again, **probability is essential**.

Slide 6: Engineering Application 3 – Communication Networks

In networks:

- Data packets travel from **source** to **destination**
- There is delay, loss, congestion
- We care about:
  - Probability that a packet arrives
  - Probability that delay exceeds some limit (QoS – Quality of Service)

So we model:

- Arrival of packets
- Delay (latency)
- Throughput

All these are **random** → we need probability theory.

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Three main parts:

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The slides you shared mainly cover **Part 1: Foundations**.

Part 1: Introduction to Probability Theory

Slide 9: Experiment

Definition:
An **experiment (E)** is a **procedure** we perform that produces some result.

Example:
Tossing a coin 5 times → call this experiment E_5

Important:

- Experiment = the **process**
- Not a single result, but the action that generates results

Slide 10: Outcome

Definition:

An **outcome (ξ)** is a **possible result** of an experiment.

Example:

For 5 coin tosses, one outcome is:

$\xi_1 = HTHTH$

So:

- Experiment: Toss coin 5 times
- Outcome: One specific sequence like HHTHT, TTTTT, etc.

Slide 11: Event

Definition:

An **event** is a **set of outcomes** of an experiment.

Example:

Let C be the event:

"Number of heads is even" in 5 tosses.

Then C includes all outcomes with:

- 0 heads
- 2 heads
- 4 heads

So:

Event = collection of outcomes satisfying a condition.

Slide 12: Sample Space

Definition:

The **sample space (S)** is the set of **all possible outcomes** of an experiment.

Properties:

- **Mutually exclusive**: You get only one outcome at a time
- **Collectively exhaustive**: All possible outcomes are included

Example:

Flip one coin:

$S = \{H, T\}$

Slides 13–14: Types of Sample Spaces

The sample space S can be:

1. **Discrete**
   - Finite or countably infinite

     Examples:
   - One coin toss: {H, T}
   - One die roll: {1,2,3,4,5,6}
   - Flip coin until first tail: {T, HT, HHT, HHHT, ...} (countably infinite)
2. **Continuous**
   - Values come from a continuous interval

     Examples:
   - Random number in [0, 1]
   - Time taken by a process

Examples listed in slides:

- Flipping a fair coin once → Discrete
- Rolling a die → Discrete
- Rolling two dice → Discrete
- Flip coin until tails occurs → Discrete, countably infinite
- Random number in [0,1] → Continuous

Slide 15: Probability and Axioms

What is Probability?

Definition:

Probability is a number that measures how likely an event is.

Mathematically:

It is a function $\Pr(A)$ that assigns a number to each event A.

Axioms of Probability

These are basic rules (no proof needed):

Axiom 1:
For any event A:
$0 \leqslant \Pr(A) \leqslant 1$

Meaning: Probability is always between 0 and 1.

Axiom 2:
If S is the sample space:
$\Pr(S) = 1$

Meaning: Something in the sample space **must happen**.

Axiom 3 (Additivity):

(a) If $A \cap B = \varnothing$ (disjoint):
$\Pr(A \cup B) = \Pr(A) + \Pr(B)$

(b) For infinitely many mutually exclusive events $A_1, A_2, \ldots$:
$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$

Meaning: If events cannot occur together, the probability of "any one of them happening" is the **sum** of their probabilities.

Slides 16–17: Corollary from Axioms

For a **finite** number M of mutually exclusive events $A_1, A_2, \ldots, A_M$:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

Explanation:
This is just the **finite version** of Axiom 3.

Note from slide:

- Axiom 3 is especially important when the sample space is **infinite**.

Slides 18–20: Propositions from Probability Axioms

Proposition 2.1: Complement Rule

$$\Pr(A^c) = 1 - \Pr(A)$$

Logic / Derivation:

- Either A happens or A^c happens
- Together they cover the whole sample space S

So:

$$\Pr(A) + \Pr(A^c) = \Pr(S) = 1$$

$$\Rightarrow \Pr(A^c) = 1 - \Pr(A)$$

Proposition 2.2: Monotonicity

If $A \subset B$, then:

$$\Pr(A) \leqslant \Pr(B)$$

Logic:

- A is contained inside B
- B has all outcomes of A and possibly more
- So B cannot be less likely than A

Conceptual + Mathematical Picture

Conceptually:

- Experiment = process
- Outcome = one result
- Event = set of results
- Sample space = all results
- Probability = how likely an event is

Mathematically:

- Events are sets
- S is the universal set
- $\Pr(\cdot)$ is a function: Events $\to$ [0, 1]
- Axioms define how this function behaves

Summary of This Part of the Lecture

You learned:

- Experiment, Outcome, Event, Sample Space
- Types of sample spaces: Discrete and Continuous
- Meaning of probability
- Three axioms of probability
- Corollary for finite mutually exclusive events
- Two key propositions:
  - $\Pr(A^c) = 1 - \Pr(A)$
  - If $A \subset B$, then $\Pr(A) \leqslant \Pr(B)$

These form the **foundation** for:

- Joint Probability
- Conditional Probability

Formula Sheet (From These Slides)

- $0 \leqslant \Pr(A) \leqslant 1$
- $\Pr(S) = 1$
- If $A \cap B = \varnothing$:
  $\Pr(A \cup B) = \Pr(A) + \Pr(B)$
- For mutually exclusive A_i:
  $\Pr\left(\bigcup_i A_i\right) = \sum_i \Pr(A_i)$
- Complement:
  $\Pr(A^c) = 1 - \Pr(A)$
- If $A \subset B$:
  $\Pr(A) \leqslant \Pr(B)$

If you want, I can now continue **in the same lecture-scribe style** for the **Joint Probability** and **Conditional Probability** parts (card deck, costume party, poker, missing key, etc.).